

Delay/Disruption Tolerant Networking  
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Probabilistic Contact Metadata for DTN Bundle Routing  
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Abstract

This document defines a generic uncertainty propagation layer for Delay/Disruption Tolerant Networking (DTN) routing: the Contact Probability Block (CPB), a BPv7 extension block that carries probability metadata in-bundle between forwarding nodes. The contribution is the standardized transport mechanism rather than any single routing algorithm; CPB is designed to be consumed and produced by heterogeneous DTN routing protocols (PROPHET, CGR, MaxProp, RAPID, Spray-and-Wait). Integrating CPB is an optional consumer/producer change at the routing agent; the block does not replace those algorithms.

The specification uses CBOR floating-point encoding in bundle extension blocks, with half-precision (IEEE 754 binary16) recommended for bandwidth efficiency. The extension preserves endpoint identifier immutability, is backwards-compatible with BPv7, and integrates with existing security mechanisms (BPsec). Distinct probability semantics from different routing families are kept separable via an explicit metric-type tag; cross-metric arithmetic is prohibited.

This document is Experimental. Its contribution is the standardized BPv7 extension block for carrying probability metadata, not a claim of routing-performance improvement. The first-draft validation covers the specified CBOR format and reference implementation. Operational deployment questions – including robustness to noisy or adversarial estimates, control-theoretic stability under feedback, and interoperability across independent BPv7 implementations – remain open.

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## 1. Introduction

Delay-tolerant networks (DTNs) often face uncertain contact schedules, such as during Mars mission disruptions, where standard BPv7 routing like Contact Graph Routing (CGR) may delay delivery due to reliance on fixed contact plans. Probabilistic routing metadata enables adaptive forwarding decisions by communicating contact likelihood estimates between nodes.

This specification defines a bundle extension block (the Contact Probability Block, or CPB) for carrying probabilistic contact metadata, enabling efficient routing decisions in DTNs for applications like space exploration communications.

### 1.1. Problem Statement (IPNSIG SSI Architecture, Section 4.2)

The Interplanetary Networking Special Interest Group (IPNSIG), the Interplanetary Chapter of the Internet Society, published "Solar System Internet Architecture and Governance – from the Moon to Mars and beyond" in September 2023 [IPNSIG2023]. Section 4.2 ("Routing and Forwarding") of that document frames the routing problem this specification addresses:

```
| The routing and forwarding procedures of the terrestrial Internet
| would not adapt well in the Solar System Internet, as topological
| state information could not be provided in a timely manner due to
| the lengthy signal propagation delays, frequent lapses of
| connectivity, and dynamic topology... New technology is needed.
```

The first "Considerations" bullet of IPNSIG Section 4.2 directly invites the design approach taken by this specification:

```
| By what delay-tolerant mechanism(s) does this technology obtain
| the information on which it decides which next-hop node to forward
| a given bundle to? Does it do so by route computation, by scoring
| and/or preferential ranking of neighboring nodes, or in some other
| way?
```

The Contact Probability Block defined here is precisely such a "scoring and/or preferential ranking" signal, carried in-bundle as metadata for forwarding nodes to consult. The CPB design responds directly to two of IPNSIG Section 4.2's four numbered Recommendations:

Recommendation 1 (Autonomy and Automation): "routing and forwarding technologies are needed that do not require continuous human intervention or management from Earth." CPB enables autonomous, in-bundle probability annotation that each forwarder consults locally, without round-trips to Earth.

Recommendation 2 (Standards): "Autonomy, in turn, requires standard methods of: propagating whatever information is needed to support route computation; utilizing that information in common ways that assure coherent forwarding." CPB standardizes both the wire format for propagating probability information (Section 3.4) and the precedence and matching rules for utilizing it (Section 3.6, Section 4.2).

Section 5 positions CPB relative to the routing-protocol families catalogued in IPNSIG Section 4.2 and its Appendix A: PROPHET [RFC6693], Spray and Wait [Spyropoulos05], Schedule-Aware Bundle Routing (SABR) [SABR], Contact Graph Routing (CGR) [CGR], and Opportunistic CGR with Uncertain Contact Plans (CGR-UCoP) [Fraire18]. CPB is a cross-protocol metadata layer, not a replacement for any of these algorithms; each can consume or produce CPB data.

## 1.2. Motivation

Standard DTN routing protocols operate with deterministic contact schedules or historical encounter statistics. However, in highly disrupted networks:

- \* Contact schedules change unpredictably (dust storms on Mars, solar weather, orbital variations)
- \* Historical data becomes stale

- \* Coordinated path computation elements (PCEs) can estimate probabilities but need a way to communicate them to forwarding nodes
- \* Adaptive routing decisions require explicit probability metadata at the bundle level

By embedding probability estimates directly in bundles as a standard extension block, this specification enables all DTN routing protocols to benefit from more informed forwarding decisions.

### 1.3. Requirements Language

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "NOT RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in BCP 14 [RFC2119] [RFC8174] when, and only when, they appear in all capitals, as shown here.

### 1.4. Terminology

**Contact Probability** The likelihood of successful bundle delivery to a destination node via a particular next hop or path, expressed as a value in [0.0, 1.0].

**Contact Probability Block (CPB)** A bundle extension block that carries contact probability metadata, routing hints, and temporal context.

**Path Probability** A probability estimate specific to a particular next-hop neighbor or outgoing interface.

**Default Probability** An overall or baseline probability estimate when per-path probabilities are unavailable.

**Metric Type** An identifier distinguishing the semantic meaning of the probability value (e.g., PROPHET delivery predictability, CGR contact confidence, MaxProp path reliability).

**Validity Duration** The time period (in seconds) after creation during which the probability estimate remains usable for routing decisions.

### 1.5. Notation Conventions

This document uses three distinct notations for describing data structures, following the conventions of other DTN Working Group drafts:

CDDL (Concise Data Definition Language) Formal schema definitions per [RFC8610]. Listings are labeled "type: cddl". CDDL defines the structural constraints that implementations MUST conform to.

CBOR Extended Diagnostic Notation (EDN) Human-readable representations of CBOR data per [RFC8610] Appendix G. Listings are labeled "type: cbor-diag". EDN examples show concrete data values in readable form.

Annotated Hex Binary Base-16 encoded wire format with inline comments. Listings are labeled "type: hex". These show the exact byte sequences that appear on the wire.

## 2. Architectural Considerations

This section explains the architectural rationale for using bundle extension blocks rather than URI query parameters for probabilistic routing metadata.

### 2.1. Why NOT URI Query Parameters

An earlier design considered encoding probability as a URI query parameter in the destination EID:

```
<CODE BEGINS>
ipn:100.1?prob=0.75
<CODE ENDS>
```

That construction is architecturally flawed: it violates fundamental DTN principles for the following reasons:

#### 2.1.1. Endpoint Identifier Immutability

Endpoint identifiers (EIDs) are immutable addresses that uniquely identify bundle destinations. If routing metadata were embedded in EIDs:

- \* Intermediate routers would need to modify destination addresses to update or strip probability parameters
- \* This violates the end-to-end principle and the Bundle Protocol specification [RFC9171]
- \* Routers cannot determine whether ipn:100.1?prob=0.75 and ipn:100.1?prob=0.5 represent the same or different endpoints
- \* Destination nodes would receive bundles with varying EID formats depending on router processing

- \* Security mechanisms binding bundles to specific EIDs (BPSec [RFC9172]) would break if routers modified destinations
- \* EID-based duplicate detection would fail when the same logical endpoint has multiple representations
- \* Custody transfer mechanisms would malfunction

#### 2.1.2. Router Modification Problem

If probability were in the destination EID, the following scenario occurs:

1. Router A receives a bundle destined to ipn:100.1?prob=0.75
2. Router A consults the probability for forwarding decisions
3. Router A must now decide:
  - \* Forward as-is with ?prob=0.75 (but this value may be stale/incorrect for the next hop)
  - \* Strip the parameter to ipn:100.1 (modifying the destination address)
  - \* Update the parameter to a new value, e.g., ?prob=0.6 (also modifying the destination address)
4. Any modification violates the immutability principle and breaks downstream systems

This architectural violation is unacceptable in the DTN model.

#### 2.2. Solution: Bundle Extension Blocks

This specification resolves the architectural conflict by carrying probabilistic metadata in bundle extension blocks, completely separate from endpoint identifiers:

- \* Destination EID remains immutable and unchanged: ipn:100.1
- \* Routing metadata travels in extension blocks (the Contact Probability Block)
- \* Routers can freely add, remove, or modify metadata without affecting the destination EID



- \* Security is maintained independently using Bundle Integrity Blocks (BIBs)
- \* Legacy nodes safely ignore unknown extension blocks without affecting bundle delivery

This design maintains clean architectural separation between:

- \* Addressing (EIDs): immutable, end-to-end
- \* Routing Hints (CPB metadata): mutable, hop-by-hop

### 3. Contact Probability Block

#### 3.1. Overview

The Contact Probability Block (CPB) is a standard BPv7 bundle extension block that carries contact probability metadata. Its purpose is to communicate probability estimates from source nodes, path computation elements (PCEs), or intermediate routers to downstream forwarding nodes.

#### 3.2. Block Structure

The CPB has the standard bundle extension block structure defined in [RFC9171]. The canonical block structure is a CBOR array per RFC 9171 Section 4.3.2:

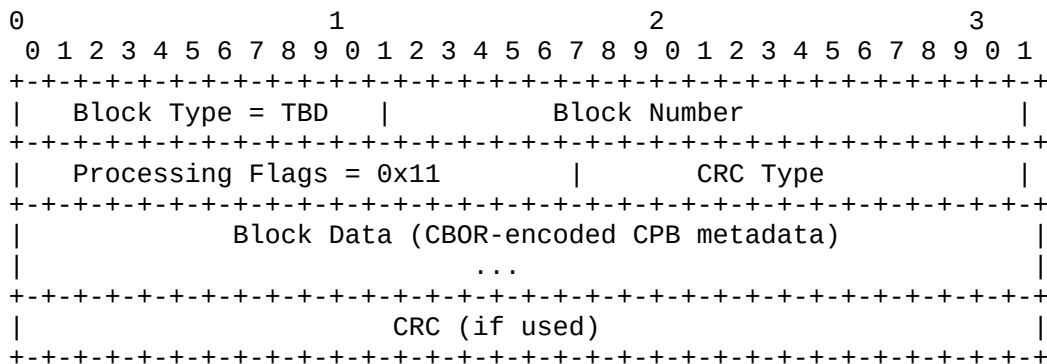


Figure 1: Bundle Extension Block Format (Octet Diagram)

The block-type-specific data (BTSD) field is a CBOR byte string (bstr) whose content is a CBOR-encoded cpb-data map. This byte-string embedding follows RFC 9171's canonical block structure, where the BTSD type is bstr (not any). The formal CDDL binding follows below.

The following annotated hex shows a concrete CPB with prob=0.75 for destination node 100:

```
<CODE BEGINS>
18 C8          ; type=200 (RFC 9171 experimental range)
02            ; num=2
11            ; flags=0x11 (replicate; discard-if-unproc)
00            ; crc-type=0 (no CRC; CRC field omitted)
57            ; BTSD: bstr (23 bytes)
  A4          ; map with 4 entries
    00 F93A00 ; 0: prob=0.75 (float16)
    01 81     ; 1: array of 1 path entry
      82     ; [node, prob]
      18 64   ; node=100
      F93C00  ; prob=1.0
    02 1A 00F73A80 ; 2: timestamp (CPB epoch seconds)
    04 19 0E10    ; 4: validity=3600s
<CODE ENDS>
```

Figure 2: CPB Concrete Example (type: hex)

The CBOR data structure in the block-data field is defined as follows. The BTSD is byte-string-embedded per RFC 9171's canonical block structure, using the CDDL socket and generic rules from RFC 9171 Section 4.3.2:

```
<CODE BEGINS>
; Integration with RFC 9171 canonical block structure.
; The BTSD is a bstr whose content is CBOR-encoded cpb-data.
; Hop-by-hop CPB MUST stay readable; do not BCB-encrypt (Sec 8.3.3).
; Plain bstr alt = RFC 9171 embedded-cbor (opaque BTSD).

; 200 = experimental-range example used in this document
; (RFC 9171 192-255). Replace after IANA assignment.
$extension-block /=
  extension-block-use<200, (bstr .cbor cpb-data / bstr)>
<CODE ENDS>
```

Figure 3: CPB Extension Block Binding (type: cddl)

```

<CODE BEGINS>
cpb-data = {
  ? 0: probability,           ; Default/overall probability
  ? 1: [* path-entry],       ; per-path (SHOULD limit size; Sec 3.4)
  ? 2: uint,                 ; Timestamp (CPB epoch seconds)
  ? 3: bstr,                 ; Source PCE identifier (optional)
  ? 4: uint,                 ; Validity duration (seconds, TTL)
  ? 5: metric-type,          ; Metric type identifier
  ? 6: probability,          ; Confidence/variance estimate
  ? 7: uint,                 ; format version (1 = this spec)
  * (int .gt 7) => any       ; Extensibility (future fields)
}

probability = float           ; See Section 3.4 for encoding

; At least one of field 0 or field 1 SHOULD be present for a
; usable estimate. An empty map is valid CBOR but no routing signal.

path-entry = [
  next-hop: eid-reference,    ; Next-hop node identifier
  probability: float          ; Probability for this path
]

eid-reference = uint / tstr   ; node number or full EID text

metric-type = &(
  prophet-dp: 0,              ; PROPHET delivery predictability
  cgr-confidence: 1,          ; CGR contact confidence
  maxprop-cost: 2,            ; MaxProp path cost (inverted)
  rapid-utility: 3,           ; RAPID utility metric
  generic: 4,                 ; Unspecified/generic probability
) / uint                      ; Extensible via future assignment
<CODE ENDS>

```

Figure 4: CPB Data Structure (type: cddl)

*\*Note on bstr .cbor embedding:* The CDDL .cbor control operator ([RFC8610] Section 3.8.4) indicates that the byte string contains valid CBOR matching the cpb-data rule. The plain bstr alternative in the extension-block binding matches the RFC 9171 embedded-cbor pattern and accommodates BTSD that is opaque under a confidentiality context applied to the enclosing block structure. For hop-by-hop routing use, CPB MUST remain readable (see Section 8.3.3); do not BCB-encrypt the CPB in that deployment mode.

*\*Field Semantics:*

Field 0: Default Probability An overall probability estimate when

per-path data is unavailable. Range [0.0, 1.0].

**Field 1: Per-Path Array** An array of [next-hop, probability] tuples. Senders SHOULD NOT include more than 8 entries. Receivers on bandwidth- or memory-constrained links SHOULD enforce a local policy limit (8 entries is RECOMMENDED) and MAY treat a CPB exceeding the local limit as malformed for that deployment, logging per [RFC9675] Section 4.3. The CDDL production uses a bound of 8 as an informative size recommendation for typical constrained-link use cases; the normative limit is the SHOULD language above. Receivers and senders on unconstrained links MAY use larger arrays.

**Field 2: Timestamp** Creation time of this CPB estimate, in whole seconds since 2000-01-01 00:00:00 UTC (the same epoch origin as DTN Time in [RFC9171], but this field is defined in \*seconds\*, not the millisecond DTN Time unit of RFC 9171 Section 4.2.6). Receivers MUST interpret field 2 and field 4 in this same second-based unit. Enables receivers to compute age and apply local decay policies.

**Field 3: Source PCE Identifier** An opaque byte string identifying the entity (e.g., a PCE or router) that computed the probability values in this CPB. This field supports trust and policy decisions (see Section 8). No particular encoding or registry is defined. The value is meaningful only within the administrative scope(s) that trust the producer. Deployments that do not require source attribution MAY omit this field.

**Field 4: Validity Duration (TTL)** The number of seconds after the timestamp during which this probability estimate remains usable. When the sum of timestamp and validity duration is less than the current time, the CPB is expired and SHOULD be ignored in full for routing (all probability fields, including per-path entries). If omitted, implementations SHOULD apply deployment-specific age thresholds (Section 4.1).

**Field 5: Metric Type** Identifies the semantic meaning of the probability value. When absent, implementations SHOULD treat the value as generic (type 4).

**Field 6: Confidence** An optional float value in [0.0, 1.0] representing the confidence or certainty in the probability estimate. A confidence of 0.9 with  $P = 0.5$  (based on 1000 observations) carries more weight than confidence of 0.1 with  $P = 0.5$  (based on 2 observations). Routing algorithms MAY use this for risk-aware forwarding.

Field 7: Version CPB format version number. If absent or set to 1, the structure and semantics are as defined in this specification. Receivers SHOULD process all recognized fields when encountering a higher version number and MAY ignore unrecognized higher fields or additional semantics defined in future specifications. Senders using only the fields defined here SHOULD omit the version field or set it to 1.

Receivers encountering an unrecognized metric-type value SHOULD ignore that CPB for routing unless local policy explicitly maps the codepoint. They MUST NOT silently coerce an unknown metric-type into metric-type 4 (generic) for automated arithmetic or ranking. Senders MUST use only registered values defined in the IANA "CPB Metric Types" registry (see Section 9) unless operating in a private or experimental context.

### 3.2.1. Block Processing Flags

The CPB MUST set the block processing control flags ([RFC9171] Section 4.2.4) as follows:

- \* \*Bit 0 (0x01): Replicate in all fragments:\* Implementers MUST set this bit so that if a bundle with a CPB is fragmented during transmission, each fragment carries a copy of the CPB.
- \* \*Bit 4 (0x10): Discard block if it can't be processed:\* Implementers MUST set this bit because the CPB is advisory routing metadata, not critical to bundle delivery.
- \* \*Bit 1 (status report if unprocessable) and bit 2 (delete bundle if unprocessable):\*MUST be 0. CPB is advisory; an unreadable CPB MUST NOT generate a status report or delete the bundle.
- \* \*Bits 3 and 5-7:\*MUST be 0 (unassigned in [RFC9171]).

In practice, both flags set means the block processing flags byte is set to 0x11 (binary 00010001).

\*Path-entry handling under fragmentation:\* When replicating a CPB into a fragment, the fragmenting node MAY prune path-entry elements (field 1) that are known to be unreachable in the fragment's remaining path, but MUST NOT alter the probability values of retained entries. The default-probability field (0) and metric-type field (5) MUST NOT be modified during fragmentation.

### 3.2.2. Cyclic Redundancy Check (CRC) Handling

CRC Type and CRC are defined in [RFC9171] Sections 4.2.1 and 4.2.2. Their placement in the canonical bundle block array is specified in Section 4.3.2. The CRC is optional:

- \* **\*No CRC (recommended):\*** Set `crc-type` to 0 ([RFC9171] Section 4.2.1) and omit the CRC field from the canonical block array (five elements total), per [RFC9171] Section 4.3.2. This is the default and minimizes overhead. Implementations **MUST NOT** emit a CBOR null CRC element when `crc-type` is 0.
- \* **\*Optional CRC:\*** Implementations **MAY** include CRC type 1 (X.25 CRC-16) or type 2 (CRC32C / Castagnoli) as defined by [RFC9171] Sections 4.2.1 and 4.2.2.

Implementations **SHOULD** prefer BPsec BIBs (Section 8.3) over CRCs for integrity, as BIBs provide both authentication and integrity.

### 3.2.3. Hop Count Block Co-Presence

Bundles carrying a CPB **SHOULD** also carry a Hop Count block (block type 10, [RFC9171] Section 4.4.3). The Hop Count block provides a loop termination guarantee independent of probability-based routing decisions.

## 3.3. Block Type Code

The bundle extension block format ([RFC9171]) requires each extension block to have a type code that identifies what kind of block it is.

### 3.3.1. IANA Registration

This specification requests IANA to assign a permanent block type code from the "Bundle Block Types" registry (created by [RFC9171]). Until IANA assignment:

- \* Implementations **MAY** use values in the 192-255 range for experimental/private use ([RFC9171] Section 4.3.2)
- \* The example hex dump uses code 200 (0xC8) for illustration
- \* Production deployments **SHOULD** use the IANA-assigned code

### 3.3.2. Backwards Compatibility

Routers that do not recognize the CPB block type code process the canonical block as specified in [RFC9171] Section 4.2.4:

1. Read the block processing control flags (the third element of the canonical block array, not a fixed octet offset).
2. If bit 4 (0x10) is set, as this specification requires, discard the unrecognized CPB and continue the bundle.
3. There is no BPv7 flag meaning "remove if can't process" (that language is from BPv6). Do not invent one.

#### 3.4. CBOR Encoding of Probability Values

The CDDL schema (Section 3.2) defines probability fields as float rather than constraining to a specific precision, because enforcing float16 in CDDL is difficult for encoders to implement in practice. Instead, the precision requirement is expressed as a prose recommendation:

Implementations SHOULD choose probability values that are exactly representable in IEEE 754 binary16 (half-precision). All values in [0.0, 1.0] at routing-grade precision (~0.001 granularity from the 10-bit mantissa) are representable in binary16. When a value is representable in float16, [RFC8949] Section 4.2.1's deterministic encoding requirement ensures it will be encoded in 3 bytes.

- \* \*Range:\* [0.0, 1.0], sufficient for all probability values
- \* \*Precision:\* Approximately 3 decimal digits, adequate for routing decisions
- \* \*Overhead:\* 3 bytes per value when float16 is used
- \* \*Standard:\* Defined in [RFC8949] (CBOR specification)

Implementations that produce probability values not representable in float16 will incur 5 bytes (float32) or 9 bytes (float64) per value under deterministic encoding. Receivers MUST accept any valid CBOR float encoding and clamp the decoded value to [0.0, 1.0].

\*Bandwidth Considerations and Array Limits:\* A CPB with per-path probabilities to multiple relays adds overhead (~4-5 bytes per entry). To mitigate DoS risks from oversized arrays on low-bandwidth or constrained links, senders SHOULD NOT produce per-path arrays larger than 8 entries. Receivers in constrained deployments SHOULD enforce a local limit (default 8) and treat excess entries as malformed for that link, logging the event per [RFC9675] Section 4.3. This guidance is normative for resource protection. The CDDL for field 1 is [\* path-entry] (unbounded); the limit of 8 is prose, not a CDDL constraint.

### 3.4.1. Invalid Float Handling

Implementations MUST handle invalid or out-of-range probability values as follows:

- \* \*NaN (0xF97E00):\* MUST be rejected as malformed.
- \* \*+/-Infinity (0xF97C00, 0xF9FC00):\* MUST be rejected as malformed.
- \* \*Negative values:\* Treat as 0.0 (invalid probability).
- \* \*Values > 1.0:\* Treat as 1.0 (certainty).
- \* \*Subnormal values:\* SHOULD be flushed to zero for embedded FPU compatibility.
- \* \*Malformed CBOR:\* Discard the CPB block and continue processing the bundle.

In all error cases, implementations SHOULD log the error (including source, timestamp, and offending value) for debugging. Malformed CBOR MUST never cause bundle deletion or administrative action; the CPB is advisory and bundle processing continues regardless.

### 3.4.2. Encoding Details

When a probability value is representable in half-precision, the CBOR encoding uses major type 7 with additional information value 25 (16-bit IEEE 754 binary16):

Major Type: 7 (floating-point or simple)  
 Additional Info: 25 (16-bit IEEE 754 float)  
 Total Bytes: 3 (tag byte 0xF9 + 2 float bytes)

Figure 5: CBOR Encoding: Major Type 7, Additional Info 25

### 3.4.3. Hex Encoding Table

| Probability | Binary16 | CBOR Hex | Bytes |
|-------------|----------|----------|-------|
| 0.0         | 0x0000   | 0xF90000 | 3     |
| 0.25        | 0x3400   | 0xF93400 | 3     |
| 0.5         | 0x3800   | 0xF93800 | 3     |
| 0.75        | 0x3A00   | 0xF93A00 | 3     |



|      |        |          |   |  |
|------|--------|----------|---|--|
| 0.95 | 0x3B9A | 0xF93B9A | 3 |  |
| 1.0  | 0x3C00 | 0xF93C00 | 3 |  |

Table 1: CBOR Encodings for Common Probability Values

### 3.5. Metric-Type Semantics and Cross-Metric Comparison

The probability values carried in a CPB are dimensionless numbers in  $[0.0, 1.0]$ , but their `_semantics_` differ substantially across routing protocols. The `metric-type` field (field 5, Section 3.2) distinguishes these semantics. This subsection specifies the constraints on combining values across metric types.

The defined metric types carry the following distinct semantics:

**metric-type 0 (prophet-dp)** PROPHET delivery predictability  $P_{\text{A,B}}$  as defined in [RFC6693] Section 2.1.2: an empirical encounter-history statistic in  $[0.0, 1.0]$ . Reflects observed past co-location behavior (with aging and optional transitivity applied `_before_` encoding), not a scheduled-contact success probability. Values in a CPB MUST already be aged at the producing node (Equation 2 of RFC 6693); receivers MUST NOT re-apply a second aging constant unless they recompute predictabilities from their own encounter base. Default probability (field 0) SHOULD be interpreted as the producer's  $P_{\text{producer, dest}}$  toward the bundle destination. When field 1 is present, each path entry for next-hop NH SHOULD carry the producer's estimate of  $P_{\text{NH, dest}}$  (the delivery predictability of that next hop toward the destination), not a repeat of  $P_{\text{producer, dest}}$  (see Section 5.1).

**metric-type 1 (cgr-confidence)** Per-contact or per-next-hop success probability against a known contact schedule (or contact-plan edge). Reflects forward-looking confidence that a planned contact occurs as scheduled (link availability, weather, pointing). Contact start/end times, one-way light time, and residual volume remain in the contact plan, not in CPB; CPB carries only the scalar confidence the plan or local estimator associates with a hop or default path.

**metric-type 2 (maxprop-cost)** Delivery-likelihood proxy derived from MaxProp-style path cost. Producers MUST map lower MaxProp cost to higher CPB value in  $[0.0, 1.0]$  using a documented local inversion (for example,  $1/(1+\text{cost})$  or a rank-normalized score). The inversion is lossy; original cost magnitude and multi-hop cost

vectors are not recoverable from the CPB value. Hop-lists used for loop avoidance and delete-ACKs remain local MaxProp state, not CPB fields.

metric-type 3 (rapid-utility) Normalized per-packet utility from a RAPID-style optimizer, scaled by the producer into [0.0, 1.0]. This is an optimization-derived score (often over delay or resource constraints), not a calibrated delivery probability. Unnormalized utilities, delay distributions, and resource inventories are out of band; if a deployment cannot normalize utility into [0.0, 1.0] honestly, it SHOULD register a new metric-type or use metric-type 4 only as advisory.

metric-type 4 (generic) Unspecified semantic. The producer makes no claim beyond "value in [0.0, 1.0]". Consumers SHOULD NOT assume any specific interpretation. Suitable for experimental social or hybrid scores (centrality, similarity) until a dedicated metric-type is registered.

\*Cross-metric arithmetic is prohibited.\* A router MUST NOT combine values from different metric types via arithmetic (addition, multiplication, weighted averaging, etc.) and treat the result as a meaningful probability. A PROPHET predictability of 0.7 and a CGR confidence of 0.7 are not interchangeable inputs to the same routing decision. Specifically:

- \* A router implementing a confidence-weighted policy that may also use rates that consumes CPB data (for example, a policy whose cost function resembles formulations in prior work such as CGR-UCoP [Fraire18]) MUST use only metric-type 1 (cgr-confidence) values, and MUST treat path-entries of other metric types as if absent.
- \* A router applying PROPHET-style transitivity rules MUST use only metric-type 0 (prophet-dp) values, and similarly ignore others.
- \* A router consuming metric-type 4 (generic) SHOULD treat the value as advisory only and MUST NOT use it as the sole input to a routing decision.

\*Mixed-metric CPBs:\* A single CPB instance carries one metric-type (field 5). A bundle MAY carry multiple CPBs with different metric types simultaneously (for example, a CPB with cgr-confidence from a PCE and a separate CPB with prophet-dp accumulated at intermediate hops). The precedence rules of Section 3.6 apply within a metric type; across metric types, each CPB is consumed only by routers implementing that metric's algorithm. A router that does not understand a particular metric type MUST ignore that CPB and continue processing.

*\*Metric-type registry policy:* Future routing protocols defining new probability semantics SHOULD register a new metric-type code (see Section 9) rather than overloading an existing one. The CPB Metric Types sub-registry exists precisely to keep semantically distinct values separable on the wire.

*\*Calibration and normalization:* This document does not specify normalization between metric types because cross-metric equivalence has no rigorous theoretical foundation in the published DTN literature. Implementations that wish to combine evidence from multiple metrics for ranking purposes SHOULD do so using a documented, deployment-specific policy and SHOULD NOT rely on numerical comparability across types. This remains an open research question.

### 3.6. Multiple CPBs in One Bundle

A bundle may accumulate multiple Contact Probability Blocks as it traverses the network. When a bundle contains multiple CPBs with conflicting probabilities for the same destination, implementations SHOULD use the following precedence:

1. Most recent timestamp (highest value)
2. If timestamps equal, trust the source (implementation-defined trust policy)
3. If timestamps are equal AND sources are equally trusted, prefer the CPB with the more specific metric-type (e.g., cgr-confidence over generic) for the current routing protocol
4. Default behavior: use the last encountered CPB (highest bundle block number)

Implementations MUST NOT fail or misroute when encountering multiple CPBs.

```
<CODE BEGINS>
/ CBOR Extended Diagnostic Notation (EDN) /
{
  0: 0.75,                / Default probability /
  1: [
    [300, 0.5],           / To relay node 300: 0.5 /
    [100, 1.0]            / To orbiter node 100: 1.0 /
  ],
  5: 1                    / metric-type: cgr-confidence /
}
<CODE ENDS>
```

Figure 6: CPB with Per-Path Probabilities (type: cbor-diag)

```

<CODE BEGINS>
A3                ; map with 3 entries
  00 F93A00        ; key 0: default prob = 0.75
  01 82            ; key 1: array with 2 path entries
    82            ; path entry [node, prob]
    19 012C F93800 ;   node 300, prob 0.5
    18 64 F93C00   ;   node 100, prob 1.0
  05 01           ; key 5: metric-type = cgr-confidence
<CODE ENDS>

```

Figure 7: CPB Per-Path Wire Encoding (type: hex)

### 3.6.1. Per-Path Matching Rules

When a router consults the per-path array:

1. Match the next-hop node/EID against path-entry keys
2. If exact match found, use that probability
3. If no match found, use default probability (field 0)
4. If no default probability is defined, treat the CPB as providing no default; fall back to local routing state (do not invent a universal 0.5)

EID-references in the path-entry array MUST match the destination bundle's URI scheme. For IPN bundles, node numbers are encoded as CBOR unsigned integers (per [RFC9758]'s updated ipn URI scheme); for other schemes, encode the full EID as CBOR text string.

*\*Scheme Mismatch Handling:* When the next-hop scheme does not match the CPB's path-entry scheme, implementations fall back to the default probability (field 0); if no default is defined, treat the estimate as absent and use local routing state. Implementations SHOULD log the scheme mismatch for OAM correlation per [RFC9675] Section 4.3.

## 4. Operational Semantics

### 4.1. Creation

The Contact Probability Block is created by:

- \* The source application (using historical or predicted contact data)

- \* A path computation element (PCE) (using network-wide routing state)
- \* The first-hop router (using local contact statistics)

Implementations SHOULD include a timestamp (field 2) and a validity duration (field 4). Timestamps MUST be expressed in whole seconds since 2000-01-01 00:00:00 UTC (same epoch origin as DTN Time in [RFC9171] Section 4.2.6, but in seconds rather than the millisecond DTN Time unit; see field 2). Field 4 uses the same second-based unit so that timestamp + validity comparisons are dimensionally consistent.

In networks with significant propagation delays, clock synchronization is challenging. When validating a CPB timestamp, implementations SHOULD apply tolerance for clock drift and light-time delays:

- \* *\*Forward skew tolerance:* Apply a configurable drift threshold (default: +/-600 seconds). CPBs with timestamps > current time + drift threshold MUST be discarded.
- \* *\*Backward skew/age threshold:* CPBs with timestamps < current time - max\_age MUST be discarded (default max\_age: 86400s interplanetary, 3600s terrestrial).
- \* *\*Deployment-specific adjustment:* Networks SHOULD define explicit max\_drift and max\_age per link characteristics.

#### 4.2. Processing by Intermediate Routers

When a router processes a bundle containing a CPB:

1. *\*Read:* Consult the CPB's default or per-path probability values.
2. *\*Decide:* Use the probability as an input to routing decisions.
3. *\*Update (optional):* If the router has fresher probability data, it SHOULD append a new CPB block to the bundle (preserves any end-to-end integrity signatures protecting the original block).
4. *\*Forward:* Transmit the bundle with the CPB(s) to the next hop.

#### 4.3. Usage in Routing Algorithms

Routing algorithms MAY use the CPB in several ways:

- \* **\*Local state preference:** Nodes that maintain native protocol state (PROPHET RIB, CGR contact plan, MaxProp tables) SHOULD prefer that local state over foreign CPB values unless local policy explicitly imports CPB as authoritative for a domain or metric-type.
- \* **\*Threshold-based:** Forward if  $\text{prob} > \text{threshold}$
- \* **\*Weighted selection:** Choose proportionally to probability
- \* **\*Replication target ranking:** When a local spray/epidemic policy already owns a remaining copy count  $L$ , CPB probabilities MAY rank candidate relays. CPB does not encode  $L$  and MUST NOT be read as defining fanout as  $\text{ceil}(1/\text{prob})$  (see Section 5.4).
- \* **\*Priority:** Prioritize higher-probability bundles during congestion
- \* **\*Risk-aware:** Weight decisions by both probability and confidence (field 6)

#### 4.4. Stability Considerations and Feedback Dynamics

Probability-aware forwarding creates feedback between routing decisions and observed network behavior. Naive use of CPB data can produce instability that confidence-blind routing avoids. Three dynamics are particularly relevant:

**Herding** When multiple sources receive identical high-confidence CPB advertisements for the same relay, they all forward to that relay simultaneously. The relay congests, its actual delivery probability drops, but in-flight CPBs continue advertising the stale high confidence. Traffic remains herded until updated CPBs propagate, which in DTN timeframes can take hours.

**Route oscillation** If a router switches preferences whenever CPB values cross a threshold, small probability fluctuations near the threshold cause traffic to oscillate between routes. Each switch may itself generate new CPB observations that further perturb the values.

**Synchronized congestion collapse** In dense topologies, simultaneous reaction by many routers to the same CPB update can produce correlated congestion events. This is structurally similar to TCP synchronization in IP networks but with much longer recovery cycles due to DTN propagation delays.

Implementations SHOULD mitigate these dynamics. The following techniques are RECOMMENDED:

- \* **Hysteresis:** Use distinct enter and exit thresholds (e.g., switch to a route at  $p > 0.75$ , switch away only at  $p < 0.60$ ). Avoids threshold-crossing chatter on small fluctuations.
- \* **Rate limiting:** Cap the frequency of routing preference changes per (source, destination) pair (e.g., at most one switch per contact opportunity, not per bundle).
- \* **Randomized tie-breaking:** When multiple routes have CPB probabilities within a configurable epsilon (default: 0.05), select randomly rather than deterministically by lowest EID. Breaks herding without sacrificing the high-probability choice.
- \* **Probability decay under load:** Maintain a local estimate of congestion on each next-hop and apply a load-aware discount to the received CPB probability before using it in routing decisions. Closes the feedback loop locally rather than waiting for upstream CPB refresh.
- \* **Bounded credulity:** Cap how heavily any single CPB can influence the local routing decision relative to the router's own observations. Specific weights are deployment-dependent.

A formal control-theoretic stability analysis of CPB-driven feedback dynamics is not provided in this document; this remains an open research question. Implementations that omit all of the above mitigations SHOULD NOT be deployed in production DTNs where congestion or herding could degrade service.

## 5. Relationship to Existing DTN Routing Protocols

This section positions CPB relative to the existing DTN routing-protocol families catalogued in IPNSIG Section 4.2 and its Appendix A ("Routing and Forwarding State of the Art").

### 5.1. PROPHET (RFC 6693)

PROPHET [RFC6693] is an epidemic protocol with strict pruning. Each node maintains delivery predictabilities  $P_{(A,B)}$  updated on encounters (Eq. 1), aged over time (Eq. 2), and optionally improved by transitivity via an intermediate (Eq. 3). During a contact, peers exchange a Routing Information Base (RIB) of destination-scoped predictabilities, then apply a local forwarding strategy (for example GRTR / GTMX) and queueing policy. Positive delivery acknowledgments prune residual copies.

\*What CPB carries for PROPHET:\* metric-type 0 values that a producer has already computed (aged, optionally transitive) for the bundle destination and/or candidate next hops. CPB is an optional in-bundle hint so downstream nodes that do not speak the full PROPHET RIB exchange can still see a snapshot of the producer's P.

\*What stays local (not CPB):\* aging constants ( $\gamma$ ,  $K$ ),  $P_{\text{encounter}}$  /  $\beta$  parameters, the full RIB dictionary, Hello / anti-entropy session state, bundle offer and response TLVs, custody policy, and named forwarding/queueing strategies. CPB does not replace the PROPHET control plane in RFC 6693 Sections 3-5.

## 5.2. MaxProp

MaxProp estimates delivery likelihood from encounter history, prefers low path-cost relays, prioritizes new packets by hop count, uses hop-lists to avoid cycles, and floods delete-ACKs so peers can drop delivered bundles. Buffer drop order is tightly coupled to the same likelihood metric used for forwarding.

\*What CPB carries:\* metric-type 2 as a lossy  $[0.0, 1.0]$  inversion of path cost / delivery likelihood so foreign nodes can rank next hops without MaxProp's full peer exchange.

\*What stays local:\* cost vectors, hop-lists, delete-ACK flooding, and the dual use of likelihood for queue-drop priority (though a CPB-aware node MAY use metric-type 2 values as one input to local drop ranking; see Section 5.10).

## 5.3. RAPID

RAPID treats routing as a resource allocation problem: it computes a per-packet utility (often over expected delay under bandwidth and storage constraints) and replicates in decreasing marginal-utility order. Peers may exchange delay samples or related metadata to refine utilities.

\*What CPB carries:\* metric-type 3 only after the producer normalizes utility into  $[0.0, 1.0]$ . CPB does not carry raw delay distributions, residual bandwidth, or multi-objective utility vectors.

## 5.4. Spray-and-Wait Family

Spray-and-Wait [Spyropoulos05] limits replication to  $L$  copies. Binary spraying and Spray-and-Focus variants choose relays uniformly or by a utility in the focus phase. The remaining copy count is the primary control state.



\*What CPB carries:\* optional ranking hints (metric-type 0/1/2/3/4) for `_whom_` to spray to when a node chooses non-uniform targets.

\*What stays local / out of scope for CPB:\* the integer `L` remaining-copy counter and spray-phase state. Those are replication-control metadata, not probabilities; they need a different extension or local header if standardized later. CPB does not encode `L`.

### 5.5. Epidemic and Summary-Vector Protocols

Classic epidemic routing and many DTN anti-entropy designs exchange summary vectors of buffered bundle IDs so peers can request missing copies. That inventory exchange is orthogonal to probability metadata. CPB neither replaces summary vectors nor carries bundle-set digests. A node may still attach a CPB when forwarding a copy so receivers learn the forwarder's confidence snapshot.

### 5.6. Contact Graph Routing (CGR)

Contact Graph Routing and SABR-style schedule-aware routing compute earliest-arrival (or related) routes from a contact plan: contact intervals, data rates, one-way light times, and residual volume. CGR-UCoP [Fraire18] and related work add contact-failure probabilities to that plan so cost can trade latency against reliability.

\*What CPB carries:\* metric-type 1 confidences for default or per-next-hop edges so in-bundle state can move with the traffic when contact-plan dissemination is slow or partial.

\*What stays in the contact plan:\* time windows, rates, OWLT, and residual capacity. CPB is not a substitute contact plan. The closest architectural precedent for in-band routing metadata is the CGR Extension Block [CGREB]. This document does not compare forwarding policies or report routing-performance results; CGR-UCoP is prior art for simulation, not the definition of the block.

### 5.7. Social and Hybrid Metrics

Protocols such as BubbleRap, SimBet, and related social DTN routers use centrality, community labels, or similarity rather than pure encounter probability. Those scores may be published as metric-type 4 (generic) with local documentation, or as new metric-type codepoints via the IANA sub-registry in Section 9. CPB still forbids mixing them arithmetically with prophet-dp or cgr-confidence.

### 5.8. DTNMA (RFC 9675)

[RFC9675] defines a DTN management architecture orthogonal to routing. A DTNMA agent could monitor CPB-related metrics (frequency of stale CPBs, scheme mismatches, BIB verification failures) for network health assessment.

### 5.9. Neighborhood Advertisement and Scheduled Topology

Secure Advertisement and Neighborhood Discovery (SAND) [I-D.ietf-dtn-bp-sand] is a DTN Working Group control-plane protocol. Nodes advertise local topology and short-horizon SABR-style metrics (data rate, one-way delay, bit-error ratio, and validity schedules) to topological neighbors. Those advertisements are not carried as metadata on application bundles.

CPB is complementary, not a substitute for SAND. Contact intervals, rates, OWLT, and residual volume remain in the contact plan or in SAND/SABR inputs. CPB carries optional in-bundle quality or confidence samples (metric-type tagged) along the data path when those estimates should travel with the bundle. A node MAY consume SAND metrics and CPB values together; it MUST NOT treat a SAND rate or delay advertisement as a CPB metric-type value, or vice versa.

Long-lived contact schedules and bandwidth as managed objects are modeled in Time-Variant Routing work, including the scheduled-attributes YANG model [I-D.ietf-tvr-schedule-yang] and related contact-plan YANG. Inter-domain reachability with optional time windows is the subject of the DTN Peering Protocol [I-D.taylor-dtn-dpp]. None of those documents define the CPB extension block.

### 5.10. Summary: CPB as Cross-Cutting Metadata

Across the families above, a recurring split appears:

- \* \*In-bundle scalar quality (CPB):\* a [0.0, 1.0] score with an explicit metric-type, optional per-next-hop list, timestamp, and validity.
- \* \*Control-plane state (not CPB):\* encounter RIBs, contact plans, summary vectors, copy counts, hop-lists, delete-ACKs, Hello sessions, and strategy/queue policy parameters.

The Contact Probability Block is designed as that cross-protocol scalar layer rather than a replacement for any algorithm. Algorithms MAY consume CPB for forwarding rank, replication target choice, or buffer-drop priority, and MAY produce CPB for downstream nodes, without requiring the peer to implement the producer's full protocol.

## 6. Backwards Compatibility

- \* Nodes that do not recognize the CPB block type MUST ignore it and continue processing the bundle normally
- \* The bundle's destination, source, and other standard fields are unaffected by the CPB
- \* Legacy routing protocols (CGR, PROPHET, MaxProp, RAPID, Spray-and-Wait) continue to function without the CPB
- \* The CPB is optional; bundles without it use standard routing behavior

## 7. Operational Considerations

This section collects deployment, sizing, and operational guidance for CPB. It draws from the processing rules in Section 4, the overhead analysis in Section 10, and the backwards-compatibility rules in Section 6.

### 7.1. When and Where to Emit CPBs

A CPB may be created by the source application (using historical or predicted contact data), a path computation element (PCE) (using network-wide state), or the first-hop router (using local contact statistics). Deployments SHOULD document their policy for which entity is authoritative for a given bundle class.

In environments with significant propagation delay (e.g., interplanetary), a PCE close to the source is often preferable so that the initial CPB reflects the most current available model. In low-delay terrestrial meshes, the first-hop router may have fresher local observations.

## 7.2. Bundle Store and Memory Sizing

If the strip-and-replace optimization in Section 3.6 is not used, bundles can accumulate multiple CPBs across hops. Each CPB occupies approximately 8-32 bytes of CBOR plus any BPsec BIB overhead. Implementations on memory-constrained nodes SHOULD bound the number of retained CPBs per bundle and apply the optional removal of earlier instances when appending a fresher CPB.

Bundle stores SHOULD account for the additional per-bundle overhead when dimensioning for CPB-enabled traffic.

## 7.3. Computational and Storage Overhead on Constrained Nodes

Decoding a CPB map (up to 8 path entries) requires CBOR parsing of roughly 32-100 bytes. On commodity hardware this is sub-millisecond; on microcontrollers using software floating-point it may be several milliseconds per CPB.

If the CPB is protected by a BPsec BIB, each consumption requires a signature or MAC verification. Implementations SHOULD cache verification results for the duration of bundle processing.

Route recomputation triggered by new CPB observations can be bounded by the hysteresis and rate-limiting techniques described in Section 4.1.

Implementations SHOULD publish concrete profiling numbers (CPU cycles per CPB, memory delta, etc.) alongside interoperability reports, per the guidance in [RFC7942].

## 7.4. Logging, Monitoring, and OAM

Nodes SHOULD log CPB-related events (reception of invalid floats, scheme mismatches, version numbers higher than supported, BIB verification failures on CPBs, etc.) using the mechanisms defined in [RFC9675] Section 4.3.

When a CPB is discarded due to age, future-dated timestamp, or malformed content, a management event SHOULD be generated to aid debugging of clock skew or misbehaving producers.

### 7.5. Partial Deployment

CPB is designed for incremental deployment. Nodes that do not recognize the block type pass it through unchanged (see Section 6). However, the routing quality improvement is only realized by CPB-aware forwarders. Deployments SHOULD monitor the fraction of CPB-aware nodes and may choose to emit CPBs only on segments where a useful number of consumers exist.

### 7.6. Congestion and Storage Pressure

Probability-aware routing tends to concentrate traffic on high-confidence routes. Under sustained load this can lead to congestion on preferred paths and storage pressure at intermediate nodes.

The mitigations in Section 4.1 (hysteresis on route changes, minimum interval between CPB-triggered recomputations, and explicit rate limiting) are RECOMMENDED. Operators SHOULD monitor queue depths and delivery ratios on high-confidence routes and be prepared to fall back to confidence-blind policies or to reduce CPB production rate during overload.

### 7.7. Clock Considerations

CPB timestamps (field 2) and validity durations (field 4) are in whole seconds since the DTN Epoch origin (see field 2; not RFC 9171 millisecond DTN Time). In networks with light-time delay or unsynchronized clocks, the age and future-dated checks in Section 4.1 must use deployment-specific drift and max\_age values. Interplanetary links commonly use larger tolerances (e.g., +/- 600 s drift, 86400 s max age) than terrestrial ones.

## 8. Security Considerations

### 8.1. Threat Model

- \* **\*Untrusted networks:** An attacker could inject false probability values to cause mis-routing (classic blackhole/sinkhole attraction via `_inflated_` metrics that draw traffic then drop it; selfish avoidance via deflated metrics)
- \* **\*Stale data:** A captured bundle with outdated probabilities could cause poor routing decisions
- \* **\*Inference attacks:** Observing probability values over time could reveal traffic patterns or network topology

- \* *\*Route poisoning by semi-trusted relays:*\* A relay that holds valid credentials within an administrative domain may still produce CPB advertisements that systematically bias routing decisions toward itself, away from a competitor, or against specific destinations
- \* *\*Replay of stale CPBs:*\* An attacker captures a recent valid (BIB-signed) CPB and replays it later when the underlying conditions have changed
- \* *\*Inter-domain trust escalation:*\* A CPB created by an entity trusted in one administrative domain crosses into a second domain where the producer is unknown; the second domain has no basis for trust evaluation

## 8.2. Trust Model

This document does not mandate a single trust model because DTN deployments span a wide range of administrative scenarios – from a single-operator interplanetary relay network to multi-party terrestrial mesh deployments. Instead, this subsection identifies the trust questions every CPB deployment **MUST** answer in its local trust policy, and provides default guidance.

### 8.2.1. Who is Authorized to Append CPBs

By default, any node along the bundle's path **MAY** append a CPB. This is permissive and suitable for low-stakes deployments where the cost of bad routing is recoverable. For higher-stakes deployments, the local trust policy **SHOULD** restrict CPB authorship to a named set of authorized PCEs (identified by the source PCE identifier field, field 3), or to nodes possessing a specific BPSec credential. CPBs from unauthorized sources **SHOULD** be discarded, with the event logged per [RFC9675] Section 4.3.

### 8.2.2. Cryptographic Adjudication of Conflicting CPBs

When a bundle carries multiple CPBs from different sources with conflicting probabilities, the precedence rules of Section 3.6 select one. Those rules **SHOULD** be evaluated only after BIB verification of each CPB; unsigned or BIB-invalid CPBs **MUST** be excluded from the precedence calculation in trust-policy contexts that require authentication. A receiver **MUST NOT** apply the metric-type-specificity tie-breaker (Section 3.6 step 3) to elevate a less-trusted source over a more-trusted one; trust ranking takes precedence over metric specificity.

### 8.2.3. Downstream Handling of Upstream BIBs

A downstream router that cannot verify an upstream BIB (missing key material, expired credentials, unrecognized security context) has two acceptable behaviors:

- \* **\*Strict:** Treat the unverifiable CPB as absent for routing decisions, and forward the bundle unchanged so that downstream routers with the correct credentials may still consume it. This is RECOMMENDED as the default.
- \* **\*Permissive:** Use the unverified CPB advisably, treating it as metric-type 4 (generic) regardless of its declared metric-type. This is NOT RECOMMENDED in adversarial environments but may be appropriate in cooperative single-operator deployments where credential rollout lags CPB adoption.

A router MUST NOT silently strip an unverifiable BIB-protected CPB and re-sign it with its own credentials. Doing so launders provenance and breaks the append-only trust chain. If a router needs to express a different probability estimate, it MUST append a new CPB with its own BIB rather than modifying or replacing the original.

### 8.2.4. Route Poisoning by Semi-Trusted Relays

A relay that is trusted enough to participate in the network, but not trusted to make routing decisions for the entire domain, can still degrade routing by producing biased CPB advertisements. Defenses against this attack model are imperfect at the CPB layer. Mitigations SHOULD include:

- \* **Plausibility checks:** a receiver SHOULD compare the asserted CPB probability against locally observed delivery statistics for the same next-hop. CPBs that deviate beyond a deployment-specific tolerance (default: factor of 2) should be discounted or discarded.
- \* **Source reputation:** trust policies MAY maintain per-source reputation scores derived from historical accuracy, and weight CPBs by source reputation when consuming them. CPB itself does not specify a reputation mechanism; the DTN Management Architecture ([RFC9675]) is the appropriate vehicle.
- \* **Bounded influence (Section 4.4):** cap how much any single CPB can shift the local routing decision relative to the router's own observations.

None of these mitigations defeat a sufficiently capable adversary. They raise the cost of route poisoning and make casual misbehavior more detectable. Trust frameworks that fully close this gap (e.g., formal reputation aggregation or attestation chains) are out of scope for this document.

#### 8.2.5. Inter-Domain Trust

When a bundle carrying a CPB crosses an administrative boundary, the receiving domain has no a priori basis for trusting the CPB's producer. Inter-domain trust SHOULD be established via one of the following approaches:

- \* A bilateral or multilateral trust agreement that designates which PCE identifiers and BIB credentials from the foreign domain are accepted
- \* A gateway router at the domain boundary that consumes the foreign CPB advisarily, validates it against local observations, and (optionally) appends a domain-native CPB expressing the gateway's translated assessment
- \* A policy of discarding all CPBs at the domain boundary, with the receiving domain regenerating CPBs internally from its own observations

This document does not standardize inter-domain trust establishment. Deployments SHOULD document their inter-domain CPB policy explicitly. This remains an open question.

#### 8.2.6. Replay Resistance

The timestamp (field 2) and validity duration (field 4), combined with the skew tolerances defined in Section 4.1, provide replay resistance that scales with the configured age threshold. For stronger protection, BIB coverage must include the timestamp. An attacker who can rewrite the timestamp without invalidating the BIB defeats the mechanism. Implementations MUST ensure that the BIB security target list covers the entire CPB block.

### 8.3. Integrity Protection

For deployments in untrusted networks or where source authentication is required, the CPB MUST be protected using Bundle Integrity Blocks (BIBs) as defined in [RFC9172] (BPSec).



### 8.3.1. Source CPB with Append-Only Intermediates

The source creates a CPB and attaches a BIB protecting only that CPB's block number. Intermediate routers with fresher probability estimates append new CPB blocks without modifying the source's CPB.

### 8.3.2. Strip-and-Replace Model

As an alternative, implementations MAY use a strip-and-replace model where each forwarder validates the previous CPB, removes it, inserts a fresh CPB with its own probability estimate, and signs it with its own BIB. Deployments SHOULD choose one model per administrative domain and document the policy.

### 8.3.3. Prohibition on BCB Encryption

Implementations MUST NOT encrypt the CPB using a Block Confidentiality Block (BCB). Encrypting a routing-critical extension block renders it opaque to the forwarding nodes that need it. Integrity protection (BIB) is appropriate; confidentiality protection (BCB) is not.

## 8.4. Mitigations

- \* Signature verification before using CPB for critical routing decisions
- \* Timestamp validation (discard old or future-dated CPBs)
- \* Secondary validation against local contact statistics
- \* Conservative thresholds for critical payloads

## 8.5. Privacy Considerations

CPB metadata may reveal information about network topology and operational patterns. Per-path probability entries expose which next-hop nodes are reachable; the source PCE identifier (field 3) identifies the entity that computed the probability. Deployments in adversarial environments SHOULD consider omitting the source PCE identifier when source attribution is not required.

## 9. IANA Considerations

This document requests IANA to assign a code point from the Bundle Block Types registry ([RFC9171] Section 9.1) under the Specification Required policy ([RFC8126]). This document does not request a specific numeric value.

| Value | Description               | Reference     |
|-------|---------------------------|---------------|
| TBD   | Contact Probability Block | This document |

Table 2: Bundle Block Types registration request

Hex dumps and CDDL examples in this document use block type 200 (0xC8). That value is in the RFC 9171 experimental and private-use range (192-255) and is for illustration and laboratory interoperation with those examples only. After IANA assignment, implementations SHOULD parameterize the block-type code so the assigned value can replace 200.

This document also requests IANA to create a new "CPB Metric Types" registry in the Bundle Protocol registries group (not a sub-registry of Bundle Block Types). The registration policy is Specification Required. The value space is unsigned integers; this document assigns 0-4. Values 5-255 are unassigned for compact encodings; values above 255 are available if needed.

| Value | Description                     | Reference     |
|-------|---------------------------------|---------------|
| 0     | PROPHET Delivery Predictability | This document |
| 1     | CGR Contact Confidence          | This document |
| 2     | MaxProp Path Cost (inverted)    | This document |
| 3     | RAPID Utility Metric            | This document |
| 4     | Generic Probability             | This document |

Table 3: Initial CPB Metric Types registrations

## 10. Overhead Analysis

| Payload (bytes) | Base Bundle | CPB minimal (8 bytes) | CPB full (32 bytes) | Overhead Range |
|-----------------|-------------|-----------------------|---------------------|----------------|
| 50              | ~100        | ~108                  | ~132                | 8-32%          |
| 256             | ~306        | ~314                  | ~338                | 2.6-10.5%      |

|         |         |         |         |          |         |
|---------|---------|---------|---------|----------|---------|
| 1024    | ~1074   | ~1082   | ~1106   | 0.7-3.0% |         |
| +-----+ | +-----+ | +-----+ | +-----+ | +-----+  | +-----+ |
| 4096    | ~4146   | ~4154   | ~4178   | 0.2-0.8% |         |
| +-----+ | +-----+ | +-----+ | +-----+ | +-----+  | +-----+ |

Table 4: CPB Byte Overhead vs Payload Size

For kilobyte-scale payloads typical of DTN sensor data, overhead is below 3% even with full CPB fields.

#### 10.1. Computational and Storage Overhead

The table above quantifies only wire-byte overhead. CPB consumption imposes additional costs on forwarders that **SHOULD** be considered when sizing a CPB-enabled deployment, particularly on constrained nodes:

**CBOR decode cost** Decoding the CPB map (up to 8 path entries) requires CBOR parsing of approximately 32-100 bytes per CPB. On commodity hardware this is sub-millisecond; on constrained microcontrollers with software floating-point it may be several milliseconds per CPB.

**BIB verification cost** If the CPB is BPSec-protected, each consumption requires a signature verification. Public key verification (ECDSA, Ed25519, HMAC) costs vary by algorithm and hardware support. In the worst case (resource-constrained node, no hardware crypto), BIB verification can dominate CPB processing time. Implementations **SHOULD** cache verification results per (block-number, BIB-block-number) tuple within a single bundle's processing window.

**Route recomputation** Each new CPB observation potentially triggers route recomputation. For a confidence-weighted routing policy consuming CPB data (see CGR-UCoP [Fraire18] as prior art), the cost is dominated by the contact-graph traversal, which scales as  $O(C \log C)$  where  $C$  is the contact-plan size; CPB consumption adds a per-edge multiplication. The hysteresis and rate-limiting techniques of Section 4.4 bound the recomputation rate.

**Memory pressure from multi-CPB accumulation** If the strip-and-replace model (Section 8.3.2) is not used, bundles can accumulate multiple CPBs across hops. Each CPB occupies up to ~32 bytes plus BIB overhead. Bundle store memory pressure **SHOULD** be considered when designing CPB production policy; the optional optimization in Section 3.6 (a node appending a new CPB **MAY** remove earlier CPB instances) is relevant here.

**Aging-logic cost** Maintaining timestamp-aware probability decay

(Section 4.1) requires periodic re-evaluation of cached CPB values against the current clock. This is  $O(1)$  per cached CPB but the cache itself SHOULD have a bounded size.

On constrained DTN nodes, CPU and storage costs frequently exceed wire-byte costs in their operational impact. Concrete profiling numbers depend on the BP implementation, the cryptographic algorithms in use, and the deployment topology, and are not provided in this document. Implementations SHOULD publish their measured overhead alongside interoperability reports per [RFC7942].

## 11. Implementation Status

This section is to be removed before publishing as an RFC.

\_[Note to RFC Editor: per [RFC7942], this section may be removed before publication.]\_

This section records the status of implementations of the Contact Probability Block at the time of writing, in accordance with [RFC7942].

### Reference Python Implementation

- \* \*Organization:\* Independent (N. Perry)
- \* \*Repository:\* <https://github.com/luckyseoul/draft-perry-dtn-cpb>  
(<https://github.com/luckyseoul/draft-perry-dtn-cpb>)
- \* \*License:\* MIT License
- \* \*Modules:\* cpb.py (encoder/decoder), test\_cpb.py (conformance suite for wire encode/decode), and config1\_sim.py (a deterministic local policy-test harness).
- \* \*Coverage:\* Wire-format encode and decode of CPB block-type-specific data (CBOR map fields, float16 encoding, invalid-float handling, BTSD wrapping), byte-exact match against Figure 2 and Figure 7 and the hex table in Section 3.4.3, and acceptance of more than eight path entries consistent with the SHOULD (not MUST) path-array size guidance. Routing policies, multi-CPB precedence, per-path matching algorithms, BPsec flags/CRC handling, and intermediate-router update rules are specified in this document but are not automated unit tests in the reference encoder.
- \* \*Maturity:\* Reference implementation. Conformance tests pass on Python 3.10+ with cbor2 5.9.0. Round-trip byte stability verified across multiple encode/decode cycles.

\* \*Contact:\* nick@perrybrandsllc.com

**Local Policy-Test Coverage** The local policy tests exercise the confidence-weighted cost helper and its mapping to CPB metric-type 1 values. They are deterministic unit and bridge tests, not a comparative routing experiment.

No other implementations are known to the author at the time of writing. Reports of new implementations are encouraged.

## 12. First-Draft Validation

This document does not report comparative routing-performance results. Validation is limited to deterministic checks of the published format and reference code.

**Document build** The XML source is rendered to Internet-Draft text and HTML using xml2rfc.

**CDDL** The standalone CPB data schema in impl/cpb.cddl is compiled with a CDDL validator. It is a validation aid and mirrors the data schema in Figure 4.

**Wire format** The reference tests cover CPB CBOR encoding and decoding, the published encoding examples and hex table, invalid floating-point values, BTSD wrapping, and round trips.

**Policy and bridge checks** Local deterministic tests cover the confidence-weighted cost helper and its mapping to metric-type 1 CPB values. These checks do not establish end-to-end routing performance.

Further simulation or deployment evaluation may be reported in a later revision after its methodology and results have been independently reviewed.

## 13. Reference Implementation

The reference implementation supporting this specification consists of the following Python modules under impl/:

- \* \*cpb.py\* - CBOR encoder/decoder, implementing the CDDL schema in Section 3.2 and the encoding rules in Section 3.4.
- \* \*test\_cpb.py\* - wire-format conformance suite (encode/decode, hex tables, invalid floats, path-array SHOULD). Passes on Python 3.10+ with cbor2 5.9.0.

- \* `*config1_sim.py*` - a deterministic local policy-test harness.
- \* `*test_config1_policies.py*` - unit tests for the confidence-weighted cost formula.
- \* `*test_sim_cpb_bridge.py*` - bridge tests showing Configuration 1 confidences encode/decode as metric-type 1 CPB maps without changing confidence-weighted route cost ranking.
- \* `*test_draft_consistency.py*` - structural checks that draft and documentation claims stay aligned with shipped code.

### 13.1. Repository

Source code is published under the MIT License at:

<https://github.com/luckyseoul/draft-perry-dtn-cpb>  
(<https://github.com/luckyseoul/draft-perry-dtn-cpb>)

### 13.2. Running the First-Draft Validation

```
<CODE BEGINS>
git clone https://github.com/luckyseoul/draft-perry-dtn-cpb
cd draft-perry-dtn-cpb
pip install cbor2

python3 impl/test_cpb.py
python3 impl/test_config1_policies.py
python3 impl/test_sim_cpb_bridge.py
python3 impl/test_draft_consistency.py
cddl compile-cddl --cddl impl/cpb.cddl
<CODE ENDS>
```

Figure 8: Reproduction Commands

These checks validate the published format and local reference code; they are not an end-to-end routing-performance evaluation.

### 13.3. Relationship to Updates of BPv7

[RFC9171] (BPv7) has been updated by [RFC9713] (administrative record types registry, January 2025) and [RFC9758] (ipn URI scheme updates, May 2025) since this CPB work began. Neither update affects the wire format defined in Section 3:

- \* [RFC9713] concerns administrative record types; CPB is an extension block, not an administrative record, so RFC 9713 does not apply directly.

- \* [RFC9758] updates the "ipn" URI scheme. CPB implementations MUST encode and decode ipn EIDs per RFC 9758, including Default Allocator FQNNs. The LocalNode ipn URI form (Allocator=0, NodeNumber= $2^{32} - 1$ ) has no meaning off-node and MUST NOT appear in CPB path-entries transmitted to other nodes.
- \* RFC 9171 erratum 8376 notes that BPv7 ADU fragmentation interacts ambiguously with extension blocks. CPB's fragmentation handling is defined in Section 3.2.1.

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## Appendix A. Mars Mission Scenario

Consider a Mars rover (node 200) communicating with an Earth relay station via an orbiter (node 100):

1. *\*Initial contact:*\* Rover creates a critical status bundle with a CPB indicating default probability 0.6 (dusty conditions), metric-type 1 (cgr-confidence), validity\_duration 3600s.
2. *\*Rover-to-orbiter:*\* A confidence-weighted consumer using cost = latency / confidence prefers a later, higher-confidence contact when that cost is lower than the dusty path.
3. *\*Later contact:*\* Dust clears; the rover appends or replaces the CPB with probability 0.95 and a new timestamp.
4. *\*Downstream:*\* Relays consume the fresh CPB (after trust/BIB policy) for ranking only.

If the first CPB (P=0.6, validity 3600 s) is delayed 10 hours, field 4 has expired. Per Section 4.1, receivers SHOULD ignore that CPB for routing. This document does not define a residual decay formula after expiry.

## Appendix B. Acknowledgements

Brian Sipos provided review comments on CDDL, BTSD embedding, fragmentation pruning, identical-timestamp tie-breaking, the per-path array size limit, scheme-mismatch logging, future-dated timestamp discard, and IANA considerations. Those comments improved this draft. Remaining errors are the author's.

## Appendix C. Change Log

This section is to be removed before publishing as an RFC.

\_[RFC Editor Note: Please remove this entire section before publication.]\_

### C.1. Unsubmitted working draft (August 2026)

- \* Initial Experimental -00: Contact Probability Block wire format, metric-type registry, operational and security considerations, and IANA requests.
- \* CBOR float16 encoding with deterministic encoding rules; format version field (key 7); per-path array size guidance as SHOULD with local enforcement on constrained links.
- \* BPSec integrity models (append-only and strip-and-replace); BCB encryption of CPB prohibited for routing use.

- \* Relationship to PROPHET, MaxProp, RAPID, Spray-and-Wait, CGR, and DTNMA; IPNSIG Section 4.2 problem framing.
- \* First-draft validation for the document build, CDDL schema, wire format, and deterministic local policy checks; no comparative routing-performance results are reported.
- \* Reference implementation and reproduction commands (Section 13).
- \* Aligns with RFC 9171 errata relevant to CRC handling and fragmentation; cites [RFC9713] and [RFC9758] as updates related to BPv7 administration and the ipn scheme.
- \* CDDL, BTSD embedding, fragmentation pruning, timestamp tie-breaker, and IANA text.

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